

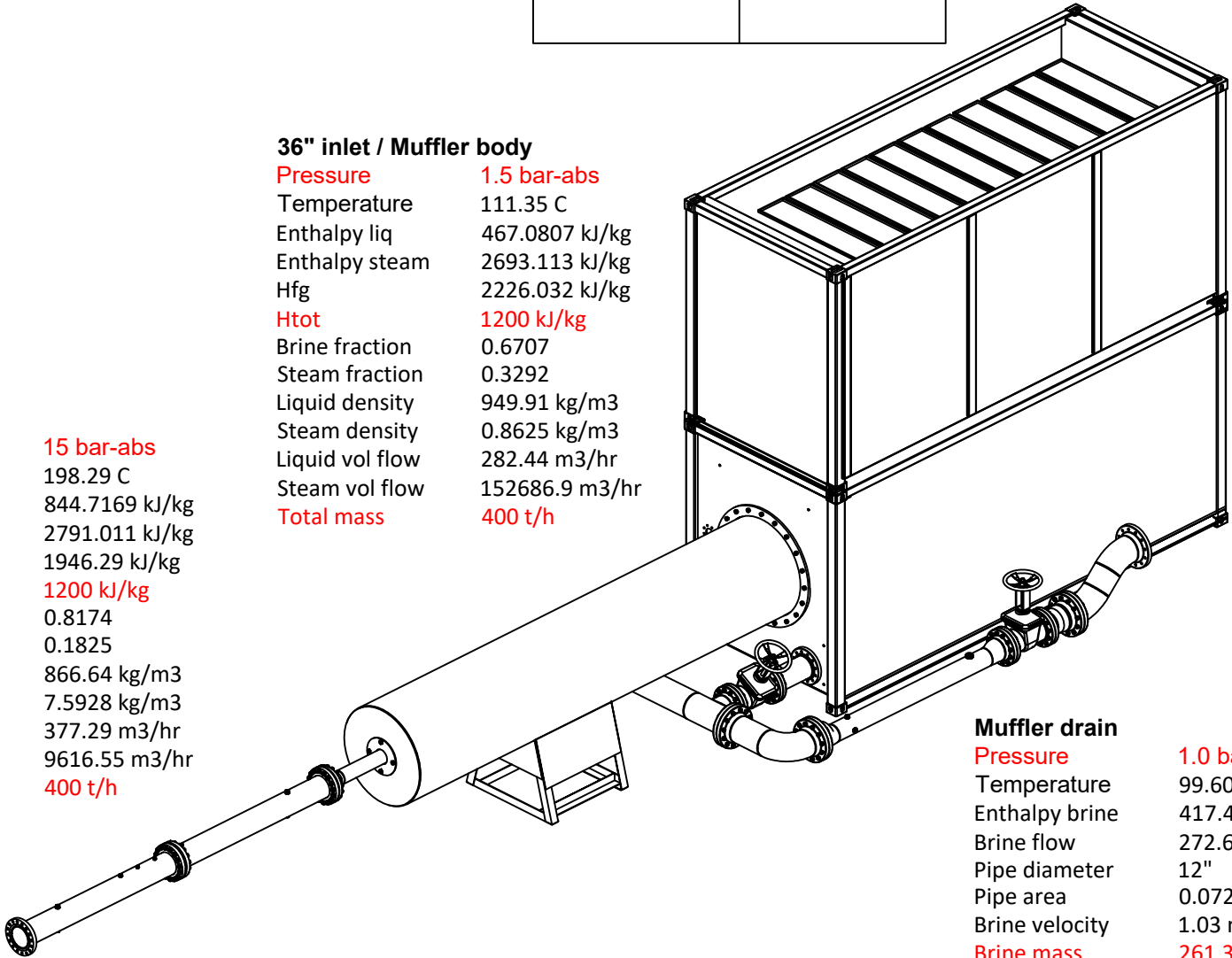
Indonesia LECM for Wapsalit and Bitung

Superficial steam velocity	
36" inlet	20' x 8' stack
m2 = 0.07296	m2 = 14.8648
m/s = 64.584	m/s = 4.38

Steam stack	
Pressure	1.0 bar-abs
Temperature	99.60 C
Enthalpy steam	2674.95 kJ/kg
Steam density	0.5903 kg/m3
Steam viscosity	1.22185E-05 Pa.s
Steam vol flow	234892.13 m3/hr
Steam velocity	4.38 m/s
Steam mass	138.65 t/h

36" inlet / Muffler body	
Pressure	1.5 bar-abs
Temperature	111.35 C
Enthalpy liq	467.0807 kJ/kg
Enthalpy steam	2693.113 kJ/kg
Hfg	2226.032 kJ/kg
Htot	1200 kJ/kg
Brine fraction	0.6707
Steam fraction	0.3292
Liquid density	949.91 kg/m3
Steam density	0.8625 kg/m3
Liquid vol flow	282.44 m3/hr
Steam vol flow	152686.9 m3/hr
Total mass	400 t/h

Flow line	
Pressure	15 bar-abs
Temperature	198.29 C
Enthalpy liq	844.7169 kJ/kg
Enthalpy steam	2791.011 kJ/kg
Hfg	1946.29 kJ/kg
Htot	1200 kJ/kg
Brine fraction	0.8174
Steam fraction	0.1825
Liquid density	866.64 kg/m3
Steam density	7.5928 kg/m3
Liquid vol flow	377.29 m3/hr
Steam vol flow	9616.55 m3/hr
Total mass	400 t/h



Muffler drain	
Pressure	1.0 bar-abs
Temperature	99.60 C
Enthalpy brine	417.43 kJ/kg
Brine flow	272.61 m3/h
Pipe diameter	12"
Pipe area	0.07296 m2
Brine velocity	1.03 m/s
Brine mass	261.34 t/h